

Overview: Genre & Core Identity

Stillpoint is a 2D side-scrolling Metroidvania with a time-slow mechanic at its core. You explore a fractured, interconnected world where time itself has been broken. The game combines precision platforming, ability-gated progression, and Souls-lite combat (telegraphed enemy attacks, checkpoints, and health management).

The tone is dark, atmospheric sci-fi/fantasy—think *Hollow Knight* meets *Celeste* with a dash of *Hyper Light Drifter*. The world is rendered in a minimalist, moody palette (purples, teals, deep blues) with procedural ambient audio and synthesized sound effects.

Core Mechanics

Movement & Platforming

- Run, jump, and wall-jump (via platform collision).
- Coyote time (6-frame leniency) – you can jump a few frames after walking off a ledge, making platforming feel responsive and fair.
- Standard Dash (Shift/X) – a short, fast horizontal burst with cooldown.
- Phase Dash (C) – an unlocked ability that lets you dash through enemies and projectiles, leaving behind a decoy Echo that distracts enemies for 2.5 seconds.

Combat

- Melee attack (Z/J) – a quick slash with a windup arc. Hitting enemies recharges your Fracture Meter (which fuels the Stillpoint ability).
- Shard Shot (V) – a ranged projectile that arcs slightly with gravity. Hold W+V to aim upward. It shatters destructible crystal walls and deals damage to enemies. It also deals double shield damage to the Crystal Sentinel.
- All enemies have telegraphed windups (crouching, exclamation marks, expanding rings) – giving you time to dodge or parry.

The Stillpoint (Signature Ability)

- Activate with Q (once unlocked in The Vault).
- Slows all enemies, projectiles, and timers to 15% speed for as long as you have Fracture Meter pips (max 3).
- Drains 1 pip every ~1 second of real-time use.
- Recharge Fracture Meter by landing melee hits on enemies.

- Immune bosses in Phase 3 – the King “sees through” the slow and counter-lunges at you.
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World & Progression

Linear Spine with Optional Branches

The world consists of 8 main areas + 1 optional lore room, connected by transitions (some requiring specific abilities):

1. The Fracture – tutorial area, teaches jumping and combat. Rewards: Phase Dash.
2. Echo Bridge – floating platforms, Phase Dash gauntlet. Optional exit to Upper Ruins.
3. Upper Ruins (optional) – lore-heavy dead end with 3 tablets and 1 easy enemy.
4. Crystal Cavern – vertical descent with destructible crystal walls. Rewards: Shard Shot.
5. The Forge – puzzle room with double-wide crystal barriers. Contains the Crystal Sentinel (new ranged enemy).
6. The Vault – rest room, no enemies. Rewards: Stillpoint ability. Exit requires Stillpoint.
7. The Rift – all-ability gauntlet with no floor. Combines jump timing, Phase Dash gaps, and enemy-blocked platforms.
8. Antechamber – pre-boss rest, final lore.
9. Boss Arena – The Fractured King. 3-phase final boss.

Checkpoints (Stillpoints)

- Glowing diamond poles scattered through each area.
- First activation heals 1 health pip and creates a respawn point.
- Respawning at a Stillpoint fully heals you and resets enemies.

Lore System

- Lore Fragments – amber diamond pickups that display a paragraph of environmental storytelling.
 - Collected lore is tracked per area; reading them builds the backstory of the Fractured King and the Stillpoints.
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Enemy Roster (Current)

Enemy	Behavior	Difficulty
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Fractured	Basic melee rusher. Windup (crouch + !), then slash. Patrols a set range.	Easy
Stutterer	Teleports every ~2 seconds, leaving decoy images. Fast, unpredictable melee attacks.	Medium
Fractured Slime	Summoned by the boss. Charges or hops toward the player. Splits into fragments on death.	Medium
Crystal Sentinel (new)	Ranged homing attacker. Floats above the ground. Fires slow, homing shards. Has a directional shield that blocks frontal melee. Shield breaks with 2 hits (1 from Shard Shot). Vulnerable to Phase Dash and behind-hits.	Hard

Boss – The Fractured King

- 3 phases based on health thresholds (100%→60%→30%).
 - Phase 1: Standard attacks (charge, slam, triple shot, projectile barrage).
 - Phase 2: Gains a shield (blocks frontal damage), starts teleporting, and summons Slimes.
 - Phase 3: Immune to Stillpoint. Gains a damage aura (tick damage if you stand too close). Uses a counter-lunge whenever you activate Stillpoint. Fires ultimate beam attacks.
 - Adaptation system – tracks whether you use melee or ranged attacks and biases its next move against your playstyle.
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Audio & Visual Design

Visual Style

- Minimalist vector art (all shapes drawn with Canvas – grey/teal/purple rectangles and diamonds).
- Parallax backgrounds with per-area deterministic procedural nebulae and silhouettes (ruins, crystals, debris).
- Subtle visual FX: screen shake, hitstop (freeze frames on impact), vignette, death fade, Stillpoint frost overlay, ability flash.

Audio (Procedural Synthesis)

- No audio files – all sounds generated via Web Audio API.

- Area-specific ambient drones that shift pitch and filter cutoff based on the room ID.
- Sound effects: jump, land, dash, phase dash, attack, hit, enemy death, shard shot, stillpoint activation, boss phase transitions, lore pickup.
- Stillpoint active – audio doesn't slow down (intentional, to keep feedback snappy).

UI & Feedback

- Health bar (6 pips) – changes color from white → yellow → red as health drops.
- Ability bar – shows Phase Dash, Shard Shot, and their cooldowns (or ??? if not unlocked).
- Fracture Meter – 3 diamond pips in the bottom-left; glows when Stillpoint is active.
- Checkpoint indicator – dot turns solid white when a Stillpoint is activated.
- Boss health bar – appears at the top during the King fight.
- Notifications – pop up for ability pickups, Stillpoint activation, lore fragments.

Stillpoint – Complete Development Roadmap (Master Plan)

This document synthesizes **everything** we've discussed: world design, movement/combat overhaul, smarter enemies, boss rework, QoL features, full-screen UI, advanced audio, and a final layer of polish (parries, wall-jumps, hazards, bestiary, and more).

All tasks are grouped into logical phases so you can build incrementally and test each part.

> **Note:** Charms / badge systems are **excluded** by request. We keep the game's identity distinct (time-fracture + dashes + Stillpoint), but borrow *only* the underlying **structural lessons** (exploration, feedback, depth) from great metroidvanias.

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Phase 0: Core UX & Quality of Life

Goal: Make the game feel polished, accessible, and full-screen from the very first boot.

#	Task	Files	Description
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0.1	**Full-screen Canvas UI**	`game.js`, `index.html`, `style.css`	Remove all HTML overlays (`#ui`, `#ability-bar`, `#boss-health`, `#controls-hint`). Render everything on the canvas: health hearts, ability icons (with cooldown rings), area name, Fracture meter, map hint, and controls hint (fades after 10s).
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0.2	**Tutorial Intro Stage**	`area.js`, `game.js`, `player.js`	Create a small isolated room (`tutorial_area`) with guided prompts: move, jump, attack (dummy), dash (barrier), phase dash (demonstrate early). Door opens after all steps. Skippable via Escape.
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0.3	**Visual Cooldown Indicators**	`player.js`, `game.js`	Draw circular cooldown rings **around the player's feet** for normal dash and around the ability icons for Phase Dash / Shard Shot. Rings fill up as cooldown recharges.
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0.4	**Persistent Save / Load**	`game.js`	Use `localStorage` to save: current area, player position, health, fracture meter, abilities unlocked, stillpoints activated, lore collected, boss state. Auto-save on Stillpoint activation, area transition, and ability pickup.
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0.5	**Quick Respawn / Pause Menu**	`game.js`	Press `Esc` → pause overlay with: *Resume*, *Return to Last Stillpoint*, *Restart Room* (resets enemies/destructibles), *Quit to Menu*. `R` key works anywhere (not just game over).
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0.6	**Accessibility Toggles**	`game.js`, `settings`	Add simple on/off toggles in the pause menu for **Screen Shake** and **Hitstop**. Wrap all shake/hitstop calls in `if (SETTINGS.screenShake)` / `if (SETTINGS.hitstop)`. Store in `localStorage`.
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Phase 1: Movement & Combat Overhaul

Goal: Give the player expressive, high-skill movement and satisfying, directional combat with defensive options.

#	Task	Files	Description
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1.1	**Directional Melee Attacks**	`player.js`, `game.js`	Add three attack directions: **Forward** (Z/J – current slash), **Up** (Z+Up/W – upward thrust), **Down** (Z+Down/S – downward slam). Downward slam **bounces** the player upward if it hits an enemy or destructible crystal (“pogo”).
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1.2	**Chained Dash + Phase Dash Parkour**	`player.js`	Allow **both dashes** to be used sequentially in mid-air (e.g., jump → normal dash → phase dash). Carry momentum between dashes. Add a small speed boost if chained within 10 frames.
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1.3	**Knockback & Aerial Juggling**	`enemy.js`, `game.js`	On hit, enemies launch **away** from the player with a small vertical velocity. If an enemy is already airborne, a second hit resets its vertical velocity (true juggle). Juggle hits grant **extra Fracture meter**.
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| 1.4 | **Combat Feedback Polish** | `game.js`, `audio.js` | Different hitstop/screen shake per attack: light (3fr / small), heavy slam (6fr / large), phase dash impact (4fr / medium). Add a **slow-mo "kill cam"** (0.3x speed for 8 frames) on the last enemy kill in a group. |

| 1.5 | **Player Parry / Deflect** | `player.js`, `game.js` | Add a 6-frame parry window at the **start** of the player's melee swing. If a hostile attack/projectile overlaps during those frames: nullify damage, reflect projectiles back at the enemy, and stun melee attackers. Gives players a defensive alternative to dodging. |

| 1.6 | **Charged Heavy Attack** | `player.js` | Holding the attack button (Z/J) for > 300ms triggers a **heavy sweep** – 2× damage, wider hitbox, increased hitstop, but slower recovery. Gives players a tactical choice: fast flurries vs. heavy hits for tanks/bosses. |

| 1.7 | **Wall Jump / Wall Slide** | `player.js` | When airborne and pressed against a wall, the player slides down slowly. Pressing jump while sliding pushes off horizontally. Enables vertical exploration without needing perfectly placed platforms. |

| 1.8 | **Dash Refund on Hit** | `game.js` | If the player lands **any** melee hit while airborne, instantly reset both the normal dash and Phase Dash cooldowns. This enables "combat flight" – chaining kills to traverse massive gaps. |

Phase 2: Enemies & Smarter AI

Goal: Enemies that challenge the new movement tools, feel intelligent, and communicate clearly.

| # | Task | Files | Description |

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| 2.1 | **Pit Avoidance (Base AI)** | `enemy.js` | All enemies ray-cast 10px ahead and 5px down. If they detect a drop (no platform/ground), they turn around **before** walking off the edge. Already partially done – extend to all enemy types. |

| 2.2 | **New Enemy: Lancer** | `enemy.js` | Fast charge attack that leaves a trail of damaging crystals. Must be parried (timed attack) or perfectly dodged. Spawns in Forge Wastes. |

| 2.3 | **New Enemy: Mage** | `enemy.js` | Hovers mid-air, fires homing projectiles. Teleports away when player gets close. Vulnerable to juggling. Spawns in Crystal Depths. |

| 2.4 | **New Enemy: Tank** | `enemy.js` | Slow, heavy, creates a damaging shockwave (radius). Weak spot on its back – forces flanking. Mini-boss variant for The Rift. |

| 2.5 | **Group Coordination & Adaptive Aggression** | `enemy.js` | If two enemies are near each other, they alternate attack timings so they can't both be parried at once. If player uses Stillpoint > 3 times in a fight, enemies gain a 30% slow-resistance (move at 70% speed during Stillpoint). |

| 2.6 | **Enemy Health Bars (Toggle)** | `enemy.js`, `game.js` | Draw a small health bar above enemies **only when damaged**. Red/green fill. Toggle on/off via a hidden setting (or always on for tough enemies). |

| 2.7 | **Enemy Windup "Ping" (Audio Cue)** | `enemy.js`, `audio.js` | When an enemy's windup transitions into the active attack frame, play a short, sharp, filtered high-frequency "ping" or

"chime". This gives deaf or audio-impaired players a clear warning and makes combat feel rhythm-game tight. |

Phase 3: Final Boss Overhaul

*Goal: A punishing, multi-phase fight that forces the player to use *all* their tools (dashes, directional attacks, parry, Stillpoint timing).*

#	Task	Files	Description
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3.1	Parry Mechanic (Boss)	`boss.js`	If the player attacks the boss from the front while he is <i>not</i> in an attack animation, he deflects it → player is stunned for 1s and takes 1 damage. Forces players to attack from behind or during recovery windows.
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3.2	Repulsion Field	`boss.js`	If the player stays within 80px of the boss for > 1 second, he emits a shockwave that <i>launches</i> the player across the room (no damage, but resets positioning and interrupts combos). Audio cue + visual ring tells you it's coming.
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3.3	Double Stillpoint Lunge	`boss.js`	When the player activates Stillpoint in Phase 2+, the boss lunges <i>twice</i> in quick succession. The second lunge <i>tracks</i> the player's dash direction (punishes panic dashing).
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3.4	Phase 3 – Desperation Mode	`boss.js`	Below 20% HP: attacks 50% faster, leaves <i>fire trails</i> on the ground (damage over time), and summons two <i>Lancer</i> adds every 15 seconds. Telegraph colours change (red = slam, blue = projectile, purple = teleport).
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3.5	Telegraph Clarity (Visual + Audio)	`boss.js`, `audio.js`	Each boss attack has a unique windup colour (drawn as expanding ring) and a distinct audio cue (e.g., low brass for slam, high chime for projectile). Skilled players can react by sound alone.
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Phase 4: World Restructure – Non-Linear Map

Goal: Break the linear chain into a web of interconnected regions, gated by abilities, with meaningful shortcuts, hazards, and secrets.

#	Task	Files	Description
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4.1	Redesign Area Layout	`area.js`	Split the world into <i>three regions</i> around a central hub ('The Vault'): • <i>Fractured Expanse</i> (blue/cold – Phase Dash focus) • <i>Crystal Depths</i> (teal/green – Shard Shot puzzle focus) • <i>Forge Wastes</i> (orange/red – combat + Stillpoint focus). Each region ends with a <i>Keystone</i> (replaces linear boss gate).
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4.2	The Vault as Hub	`area.js`, `game.js`	'The Vault' now has <i>three exits</i> , each requiring a specific ability to unlock: • Phase Dash → to Fractured Expanse • Shard Shot → to Crystal Depths • Stillpoint → to Forge Wastes. Hub also contains a merchant (optional) and a fast-travel gate.
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| 4.3 | ****Ability Gates & Blocked Paths**** | `area.js`, `game.js` | Add new transition types:
 • `requires: 'shard\_shot'` → crystal wall that must be shot.
 • `requires: 'stillpoint'` → time-locked door that only opens while Stillpoint is active.
 • `requires: 'phase\_dash'` → wide gap that requires phase dash to cross. |

| 4.4 | ****Shortcuts & One-Way Doors**** | `area.js` | Add locked doors that open from the other side (e.g., from Crystal Depths into Forge Wastes) – creates a sense of interconnectivity. Use `toX`/`toY` to place the player behind the door. |

| 4.5 | ****Fast Travel (Stillpoint Gates)**** | `game.js`, `area.js` | Add large activated Stillpoints that act as warp points. Unlock a “Gate Key” in the hub. Warp between any two activated gates from the map screen. |

| 4.6 | ****Environmental Hazards (Traps)**** | `area.js`, `game.js` | Add stationary sawblades, timed crystal spikes, and moving pendulum blades in The Rift and Forge Wastes. Hazards have `x, y, w, h, type, timer`. Player overlap = 1 damage + knockback. Forces dashes to be used defensively in platforming. |

| 4.7 | ****Dynamic Lantern / Player Aura**** | `game.js` | Draw a subtle, warm radial glow ****centered on the player**** (e.g., `RadialGradient` with soft opacity fading to 0) that slightly illuminates the dark background. Makes the world feel more tangible and less flat. |

| 4.8 | ****Secret Shimmer (Breakable Walls)**** | `game.js`, `area.js` | For destructible walls, add a very subtle intermittent shimmer – 1-2 sparkle particles (small white diamonds) appearing at random positions on the wall for 1 frame every ~5 seconds (`Math.random() < 0.002`). Trains players to look for secrets without hand-holding. |

Phase 5: Extra Depth & Polish

Goal: Add replayability, player agency, and a richer world without bloating the core loop.

#	Task	Files	Description
5.1	**Input Remapping Menu**	`input.js`, `game.js`	Add a simple canvas-drawn menu where players can rebind keys. Store bindings in `localStorage`. Defaults: arrows/WASD, Z/J attack, X dash, C phase dash, V shard shot, Q stillpoint.
5.2	**Advanced Reactive Audio**	`audio.js`	Expand the audio system: • **Spatial panning** – enemy sounds pan left/right based on screen position. • **Dynamic drone** – layers of oscillators that morph per area (already started). • **Combat sting** – a low-pass filtered noise + sub-bass pulse that syncs with enemy attack cooldowns. • **Boss phase music** – synthesised drum loop that speeds up as boss health drops.
5.3	**Currency System (Fractured Essence)**	`game.js`, `area.js`	Enemies drop blue “Essence” orbs. Spend them at a hub merchant for: • Map upgrades (reveal hidden rooms) • Temporary buffs (e.g., +1 damage for 10 rooms) • Unlockable cosmetics (optional).
5.4	**Boss Rush / Radiant Mode**	`game.js`, `boss.js`	After beating the game, unlock a **Boss Rush** mode (fight all bosses back-to-back with 3 heals). Also add a **Radiant** difficulty for the main boss (one-hit death, but gives a special achievement/lore).

| 5.5 | ****NPCs & Side Quests**** | `game.js`, `area.js` | Add 2–3 friendly NPCs in safe rooms. They give small quests:
 • “Find my lost crystal” → reward: extra Fracture pip.
 • “Defeat 5 Stutterers” → reward: map marker for a secret.
 • Dialogue appears as canvas-drawn text boxes (like the lore overlay). |

| 5.6 | ****In-Game Bestiary / Journal**** | `game.js`, `pause menu` | A journal that fills out as you defeat each enemy type – shows name, HP, and a cryptic lore line. Unlocked in the pause menu (drawn like the map overlay). Rewards completionists and grounds the world. |

| 5.7 | ****"Memory" Boss Refights (Hub)**** | `game.js`, `boss.js` | Once the final boss is defeated, a shimmering ghost appears in The Vault. Interacting lets you refight the boss (at your current power level) anytime – great for practice and speedrunners. Skips the "entering" cutscene. |

What This Plan Does *Not* Include (By Design)

- ~~✗~~ Charms / badge system (you explicitly rejected this).
- ~~✗~~ Geo / shop economy as a main progression gate (kept minimal in Phase 5.3).
- ~~✗~~ Procedural generation – all rooms are hand-crafted.
- ~~✗~~ Multiplayer / online features.

Immediate Next Step

Start with ****Phase 0.1 – Full-screen Canvas UI****.

It's the quickest win and immediately makes the game feel like a polished, finished product when played in full-screen.

Once that's done, move to ****Phase 0.2 – Tutorial Stage****, then ****Phase 1.1 – Directional Attacks****.